1	A formulation scientist prepares tablets using API, lactose and povidone. During testing, the tablets possess good hardness but fail the disintegration test. Outline the probable reason for the failure. Recommend a suitable excipient and justify your answer. (5 Marks)

Q1. Tablet fails disintegration test – 5 Marks

1. **Probable reason:** Excessive povidone concentration causes strong binding between particles.
2. Strong binding increases tablet hardness and **delays disintegration**.
3. **Recommended excipient:** Croscarmellose sodium.
4. Croscarmellose sodium rapidly **absorbs water and swells**.
5. This promotes **rapid tablet breakup and improves disintegration**.

2	A pharmaceutical company plans to manufacture chewable tablets for children. The formulation scientist must select a suitable diluent that improves patient acceptability. Based on the scenario provided, recommend a suitable diluent and justify your choice. (5 Marks)

Q2. Diluent for chewable tablets – 5 Marks

1. **Recommended diluent:** Mannitol.
2. It has a **pleasant sweet taste**.
3. It provides a **cooling sensation and good mouthfeel**.
4. It is **non-cariogenic** and suitable for children.
5. Therefore, it improves **palatability and patient acceptability**.

3	During the pre-formulation evaluation of a new oral drug candidate, two formulations exhibited similar drug content and pharmacological activity. However, one formulation dissolved rapidly in gastrointestinal fluids but showed poor membrane permeation, whereas the other crossed biological membranes efficiently despite dissolving slowly. Before finalizing the formulation strategy, the research team decided to evaluate the partition coefficient of both drug candidates. Analyse how the partition coefficient influences the observed behaviour of the two formulations and justify its significance in oral drug delivery. (5 Marks)

Q3. Partition coefficient in oral drug delivery – 5 Marks

1. **Partition coefficient** indicates the distribution of a drug between lipid and aqueous phases.
2. **Low partition coefficient:** Good aqueous solubility but generally poorer membrane permeability.

3. **High partition coefficient:** Better membrane penetration but may have poor aqueous solubility.
4. The rapidly dissolving formulation with poor membrane permeation likely has **lower lipophilicity**.
5. Therefore, an **optimum partition coefficient** is essential for good absorption and oral bioavailability.

4	(i) Two pharmaceutical companies manufacture the same paracetamol tablet. Company A uses only API and lactose, whereas Company B incorporates API, lactose, povidone, starch and magnesium stearate in its formulation. Compare the formulation strategies adopted by the two companies and determine which formulation is likely to produce a pharmaceutically superior tablet. Justify your conclusion. (3 Marks)
	(ii) Based on the above formulation, recommend one additional excipient that Company A should incorporate to improve its tablet formulation and justify your recommendation. (2 Marks)

Q4(i). Compare Company A and B – 3 Marks

1. **Company A:** API + lactose; lactose mainly acts as a **diluent**.
2. **Company B:** Povidone acts as **binder**, starch as **disintegrant**, magnesium stearate as **lubricant**, and lactose as diluent.
3. **Company B is superior** because its formulation provides better **tablet strength, disintegration and manufacturing properties**.

Q4(ii). Additional excipient – 2 Marks

1. **Recommended excipient:** Sodium starch glycolate (SSG), a **superdisintegrant**.
2. It rapidly absorbs water and swells, thereby **improving tablet disintegration and dissolution**.

5	A pharmaceutical student stated that a biosimilar can be manufactured in the same way as a generic drug because both are developed after the innovator product becomes available. Another student disagreed, explaining that biologics are produced using living systems and therefore cannot be reproduced as exact copies. Evaluate the two opinions by explaining why biosimilars are described as "highly similar" rather than "identical" to the reference biologic. (5 Marks)

5. Biosimilar vs Generic Drug – 5 Marks

1. Generic drugs are chemically synthesized and can be reproduced as identical copies of the reference drug.
2. Biosimilars are produced using living cells or biological systems, so exact replication is not possible.
3. Biological products are structurally complex and may show minor variations between batches.
4. Therefore, a biosimilar must demonstrate high similarity to the reference biologic in quality, safety and efficacy.
5. Hence, the second student's opinion is correct: biosimilars are described as “highly similar” rather than “identical.”

6	A pharmaceutical company compared two biosimilar products. • Product A: Identical amino acid sequence; comparable glycosylation profile. • Product B: Identical amino acid sequence; significantly altered glycosylation profile. A pharmaceutical company evaluated two biosimilar candidates during product development. • Product A: Identical amino acid sequence and a glycosylation profile comparable to the reference biologic. • Product B: Identical amino acid sequence but a significantly altered glycosylation profile compared with the reference biologic. The development team proposed that both products should receive regulatory approval because their amino acid sequences are identical. Evaluate the proposal and determine which product is more likely to satisfy biosimilar approval

	requirements. Justify your conclusion based on the critical quality attributes of biosimilar products. (5 Marks)

6. Evaluation of Biosimilar Products – 5 Marks

1. Identical amino acid sequence alone is **not sufficient** to establish biosimilarity.
2. **Product A** has an identical amino acid sequence and a **comparable glycosylation profile** to the reference biologic.
3. **Product B** has an identical sequence but a **significantly altered glycosylation profile**.

4. Glycosylation is a critical quality attribute that can affect **stability, activity, immunogenicity and pharmacokinetics**.
5. Therefore, **Product A is more likely to satisfy biosimilar approval requirements**.

7	A biosimilar developer successfully confirmed the primary amino acid sequence of a therapeutic protein using advanced analytical techniques. However, the regulatory review committee requested additional evidence before concluding that the biosimilar would perform identically to the reference product during long-term clinical use. Evaluate why analytical characterization alone may not completely eliminate structural uncertainty in biosimilar development. (5 Marks)

7. Why Analytical Characterization Alone May Not Eliminate Structural Uncertainty – 5 Marks

1. Analytical techniques can confirm the **primary amino acid sequence** of a therapeutic protein.
2. However, proteins have complex **higher-order structures** such as secondary, tertiary and quaternary structures.
3. Post-translational modifications such as **glycosylation** can also affect protein properties.
4. Small structural differences may influence **stability, biological activity and immunogenicity**.
5. Therefore, **clinical and functional studies** are required along with analytical characterization to establish biosimilarity.

8	(i) A pharmaceutical company planned to develop a transdermal patch for systemic drug delivery. During formulation evaluation, scientists emphasized that only molecules possessing suitable physicochemical properties could effectively permeate the skin barrier. Explain the physicochemical requirements necessary for successful transdermal drug delivery. (3 Marks)
	(ii) Based on the above evaluation, justify why Drug B is unsuitable for passive transdermal delivery. (2 Marks)

8. (i) Physicochemical Requirements for Passive Transdermal Delivery – 3 Marks

1. **Moderate molecular weight:** Small molecules, generally preferably below about 500 Da, penetrate the skin more easily.
2. **Suitable lipophilicity:** The drug should have sufficient lipid solubility to cross the stratum corneum while retaining some aqueous solubility.

3. **Adequate potency and low melting point:** The drug should be potent at a relatively low dose and preferably have a low melting point for better skin permeation.

8. (ii) Why Drug B is unsuitable for passive transdermal delivery – 2 Marks

1. Drug B likely has **unfavourable physicochemical properties**, such as high molecular size or excessive polarity.
2. These properties reduce its ability to **partition into and diffuse through the lipid-rich stratum corneum**, making passive transdermal delivery ineffective.

9	A researcher observed that a newly developed protein drug showed excellent therapeutic activity in the laboratory but lost its effectiveness during storage. Further investigation revealed exposure to high temperature and repeated shaking during transportation. Predict the factors responsible for the loss of therapeutic activity and explain how formulation development helps maintain the quality of biotherapeutics. (5 Marks)

9. Loss of Therapeutic Activity During Storage – 5 Marks

1. **High temperature** can cause protein denaturation, aggregation and degradation.
2. **Repeated shaking** can cause aggregation and structural damage to the protein.
3. Other factors such as **pH, oxidation, moisture and light** can also reduce protein stability.
4. Formulation development uses suitable **buffers, stabilizers and surfactants** to protect the protein.
5. Proper **storage conditions and packaging** help maintain stability, therapeutic activity and quality.

10	A pharmaceutical company developed two formulations for the same biotherapeutic. Formulation X increased only the apparent solubility of the drug, whereas Formulation Y increased the concentration of free dissolved drug molecules. The development team must choose one formulation for further studies. Evaluate both formulations and predict which formulation would be preferred for improving drug absorption and therapeutic performance. (5 Marks)

10. Formulation X vs Formulation Y – 5 Marks

1. Formulation X increases apparent solubility but may not increase the amount of free drug available for absorption.
2. Formulation Y increases the concentration of free drug molecules available for absorption.

3. Free drug molecules can readily partition into biological membranes and undergo absorption.
4. Therefore, Formulation Y is expected to provide better absorption and therapeutic performance.
5. Hence, Formulation Y should be preferred for further studies.

11	A biotechnology company is developing two therapeutic candidates for systemic treatment. • Drug X: Molecular weight = 420 Da, Log P = 3.2, Hydrogen bond donors = 2, Hydrogen bond acceptors = 6, stable in gastric fluid. • Drug Y: Molecular weight = 18,000 Da, highly hydrophilic, susceptible to enzymatic degradation, and loses activity on exposure to heat. Despite Drug X satisfying Lipinski's Rule of Five, the company plans an oral tablet, whereas Drug Y is formulated as a lyophilized injectable product. Assess the scientific basis for selecting different formulation approaches for Drug X and Drug Y. (5 Marks)

11. Drug X vs Drug Y – 5 Marks

1. Drug X has a molecular weight of 420 Da and satisfies Lipinski's Rule of Five, indicating suitability for oral drug delivery.
2. Its moderate molecular weight and suitable lipophilicity favour membrane permeation and oral absorption.
3. Drug Y has a very high molecular weight (18,000 Da), is highly hydrophilic and is poorly suited for membrane penetration.
4. Drug Y is also susceptible to enzymatic degradation, reducing its oral bioavailability.
5. Therefore, Drug X is suitable as an oral tablet, while Drug Y requires a lyophilized injectable formulation for effective systemic delivery.

12	(i) During formulation optimization, two batches of the same biotherapeutic showed identical drug content. However, one batch exhibited better absorption because a different solubility-enhancing excipient had been incorporated. Justify how solubility-enhancing excipients influence the performance of a biotherapeutic formulation. (3 Marks)
	(ii) Based on the above scenario, state two formulation parameters that may improve when drug solubility is enhanced. (2 Marks)

12. (i) Effect of Solubility-Enhancing Excipients – 3 Marks

1. Both batches have the same drug content, but the batch with the solubility-enhancing excipient has **higher drug solubility**.
2. Increased solubility produces a higher concentration of **free drug available for absorption**.
3. Therefore, the enhanced-solubility formulation can improve **drug absorption and therapeutic performance**.

12. (ii) Formulation Parameters Improved by Increasing Solubility – 2 Marks

1. **Dissolution rate** – increases due to improved drug solubility.
2. **Drug absorption/bioavailability** – increases because more drug is available in dissolved form.

13	A recombinant protein formulation exhibited visible particles after prolonged storage. Analytical studies confirmed that individual protein molecules had combined to form large complexes, resulting in reduced

	biological activity. The formulation scientist proposed modifying the formulation to improve its shelf life. Predict the cause of reduced therapeutic activity and justify the formulation strategies that can be employed to minimize protein aggregation and enhance shelf life. (5 Marks)

13. Protein Aggregation and Reduced Activity – 5 Marks

1. **Cause:** Protein molecules aggregate and form large complexes during prolonged storage.
2. Aggregation can alter the protein's **native structure and active sites**, reducing biological activity.
3. Use suitable **stabilizers** such as sugars/polyols to maintain protein structure.
4. Use **surfactants** to reduce protein adsorption and aggregation at interfaces.
5. Optimize **pH, temperature and ionic strength** and use suitable storage conditions to improve shelf life.

14	During formulation development, a protein solution showed significant loss of active drug even though no chemical degradation was detected. Further investigation revealed that protein molecules had adhered to the walls of the glass container. Predict the reasons for the reduction in active protein concentration and justify the use of surfactants in improving the shelf life of protein formulations. (5 Marks)

14. Protein Adsorption to Container Walls – 5 Marks

1. Protein molecules can **adsorb onto the glass container surface**, reducing the active protein concentration.
2. Adsorption can cause **protein unfolding and aggregation**, leading to loss of activity.
3. **Surfactants** compete with proteins at air–liquid and solid–liquid interfaces.
4. They reduce **surface adsorption and interfacial stress**, thereby preventing protein denaturation and aggregation.
5. Therefore, suitable surfactants such as **polysorbate 20 or polysorbate 80** can improve protein stability and shelf life.

15	A vaccine formulation was accidentally exposed to freezing temperatures during transportation. After thawing, visible particles and aggregation of the vaccine protein were observed. Recommend a suitable excipient to minimize freezing-induced aggregation in future batches and explain its protective mechanism. (5 Marks)

15. Freezing-Induced Protein Aggregation – 5 Marks

1. Freezing can cause **protein concentration and structural changes** in the unfrozen phase, promoting aggregation.
2. **Recommended excipient:** Sucrose.
3. Sucrose acts as a **cryoprotectant** and helps maintain the protein's native structure.
4. It reduces protein–protein interactions and protects against **freeze-induced denaturation and aggregation**.
5. Therefore, adding sucrose can improve **freeze–thaw stability and shelf life** of the vaccine.

16	(i) During stability studies, a protein biotherapeutic exhibited a gradual loss of biological activity due to oxidative degradation. Analyze the effect of oxidative degradation on the biological activity and therapeutic performance of a protein biotherapeutic formulation. (3 Marks)
	(ii) Select an appropriate formulation approach to improve the stability of the protein biotherapeutic and justify your choice. (2 Marks)

16. (i) Effect of Oxidative Degradation – 3 Marks

1. Oxidation can modify susceptible amino acid residues and alter the **protein structure**.
2. Structural changes can reduce **biological activity and receptor binding**.
3. Consequently, oxidative degradation reduces the **therapeutic effectiveness and stability** of the biotherapeutic.

16. (ii) Formulation Approach – 2 Marks

1. Add a suitable **antioxidant** such as methionine to reduce oxidative degradation.
2. Use **oxygen-protective packaging and suitable storage conditions** to improve protein stability and maintain biological activity.

17 A pharmaceutical company assigned the following suitability ratings for different dosage forms as provided in the below figure 'Topical formulations comparison chart".

Criterion	Ointment	Cream	Lotion	Gel
Occlusive property	High	Moderate	Low	Low
Ease of spreading	Low	High	High	Moderate
Greasiness	High	Low	Low	None

Figure: Topical formulations comparison chart

Using the matrix, rank the four dosage forms from the most suitable to the least suitable for managing dry chronic skin conditions.Justify your ranking.

(5 Marks)

17. Ranking of Topical Formulations – 5 Marks

Ranking from most suitable to least suitable for dry chronic skin conditions:

1. **Ointment – Most suitable:** High occlusive property helps retain moisture; however, it has low spreadability and high greasiness.
2. **Cream:** Moderate occlusiveness, high spreadability and low greasiness make it suitable.
3. **Gel:** Low occlusiveness and moderate spreadability; provides little moisturization for dry skin.

4. **Lotion – Least suitable:** Low occlusiveness, although it has high spreadability and low greasiness.

Conclusion: Ointment > Cream > Gel > Lotion because dry chronic skin conditions require better **occlusion and moisture retention**.

18	During stability testing, two liquid formulations failed quality evaluation. • Product A: A permanent solid mass formed at the bottom of the container and could not be redispersed by shaking. • Product B: Two distinct liquid layers remained permanently separated after storage. Analyse the root cause of each failure and explain how the observed instability affects the pharmaceutical quality of the respective formulation.

18. Liquid Formulation Instability

Product A

- **Failure:** Permanent solid mass at the bottom that cannot be redispersed.
- **Cause: Caking/sedimentation** due to particle aggregation.
- The sediment becomes compact and forms a hard cake.
- It cannot be redispersed, causing **non-uniform drug dosing** and poor product quality.

Product B

- **Failure:** Two liquid layers remain permanently separated.
- **Cause: Phase separation/cracking** of the emulsion.
- The dispersed phase separates completely and cannot be restored by shaking.
- This results in **non-uniform drug distribution and reduced pharmaceutical quality**.

19	A quality review committee made the following observations after evaluating four liquid dosage forms. • Observation I: Solutions provide rapid absorption because the active pharmaceutical ingredient (API) is already dissolved. • Observation II: Syrups are naturally self-preserving because of their high osmotic pressure. • Observation III: Flocculated suspensions settle slowly and form irreversible caking. • Observation IV: Oil-in-water emulsions are preferred for the oral delivery of unpalatable oily drugs. • Observation V: Suspensions are preferred only when the drug is completely soluble in the dispersion medium. Evaluate the observations and determine which require revision before the committee report can be approved. Justify your decision. (5 Marks)

19. Evaluation of Observations – 5 Marks

1. **Observation I – Correct:** Solutions provide rapid absorption because the API is already dissolved.
2. **Observation II – Requires revision:** Syrups may have self-preserving action due to high osmotic pressure, but this depends on sufficient sugar concentration.
3. **Observation III – Incorrect:** Flocculated suspensions settle rapidly but form a **loose, easily redispersible sediment**, not irreversible caking.
4. **Observation IV – Correct:** Oil-in-water emulsions can improve the palatability and oral delivery of unpalatable oily drugs.
5. **Observation V – Incorrect:** Suspensions are used for drugs that are **poorly soluble or insoluble** in the dispersion medium.

Therefore, Observations II, III and V require revision.

20 (i)

Excipient Category	Recorded Function
Binder/Adhesive	Promotes particle adhesion
Disintegrant	Improves powder flow during compression
Lubricant/Glidant	Prevents sticking to punches

Review the formulation record as provided in the above table and identify the incorrectly assigned function. Justify your answer.

(3 Marks)

(ii) Based on the audit findings, identify the excipient category that requires functional correction before formulation approval.

(2 Marks)

20. (i) Incorrectly Assigned Function – 3 Marks

1. **Incorrect function:** Disintegrant – “Improves powder flow during compression.”

2. A **disintegrant** helps the tablet break apart rapidly after contact with fluid.
3. **Powder flow** is mainly improved by **glidants**, while preventing sticking to punches is the function of lubricants.

(ii) Excipient Category Requiring Correction – 2 Marks

1. **Disintegrant category** requires functional correction.
2. Its correct function is to **promote rapid breakup of the tablet and improve drug dissolution**.